Lifelong Education for Neural Networks

Continual Lifelong Learning with Neural Networks: A Review

G. I. Parisi, R. Kemker, J. L. Part, C. Kanan, and S. Wermter

In Neural Networks, 2019

Presenter : Jiwon Yang

(jwyang@gist.ac.kr)

Contents

01 Introduction

02 Prerequiste : Knowledge Transfer

03 Challenge : Catastrophic Forgetting

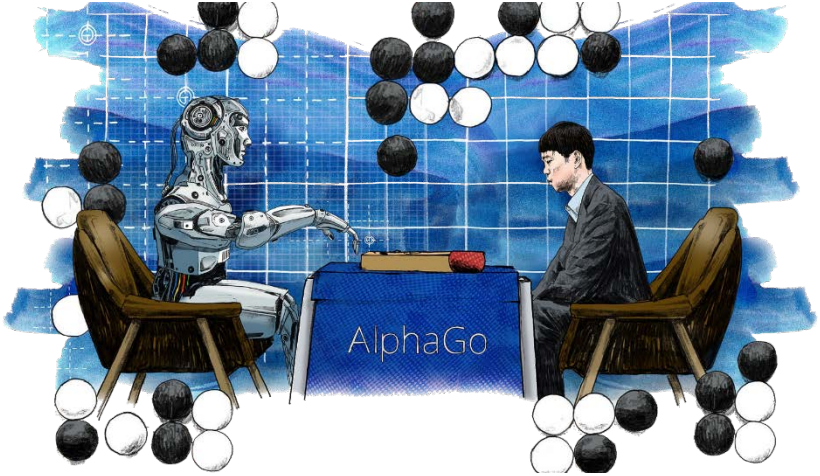
04 Solution : Continual Lifelong Learning

05 Conclusion

Advance in Artificial Intuition

* **Deep learning is more likely to surpass human-decision making**

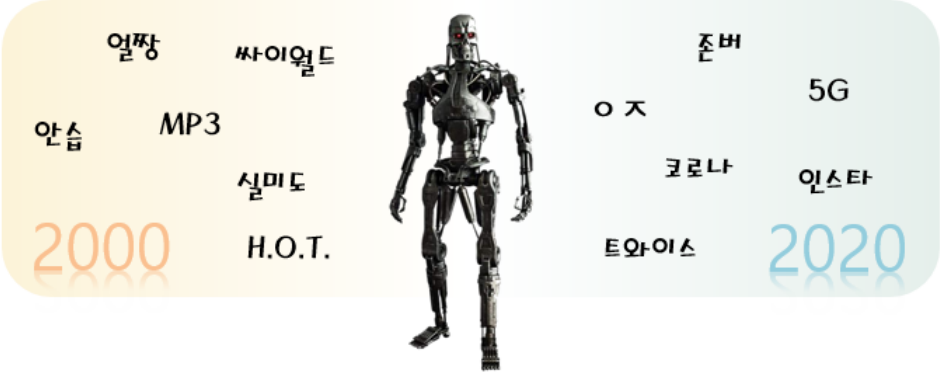
- ✓ Only Humans can make a decision based on what they feel, not necessarily what would be logical, and nowadays this becomes an old myth
- ✓ AI is not only more intelligent machines but also it seems like it has developed a form of intuition



Requirement to Reach the Perfection

* A man expands his horizons via prior experiences and know-how

- ✓ Unlike machines, humans can learn new tasks, new data by combining past knowledge without repeating memory initializations
- ✓ The current goal of deep learning is storing knowledge gained while solving one problem and applying it to a different, but related problem



▲AI also lives in continually changing enviornment

Requirement to Reach the Perfection

✓ We hope the simple Car/Pedestrian classfier also handles ...

- (New Class) → Classifying Car/Pedestrian/Bike
- (Subdivison) → Car - (Bus/Compact/Truck ...), Pedestrian - (Adult/Child ...)
- (Adaptation) → Identifying traffic sign of South Korea

✓ Conventional models lose their chance to make up for unexperienced errors after the moment one thinks all training is done

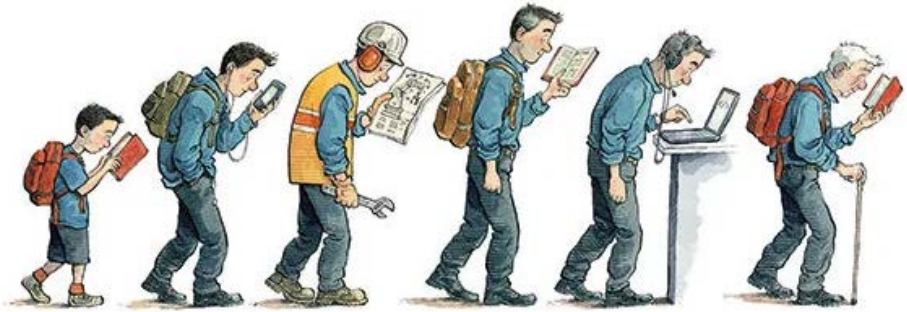
Machines also need Lifelong Learning

* Learning is a never ending process

- ✓ Lifelong learning is something that data scientists and researchers believe can further our progress towards artificial general intelligence
- ✓ The big challenge will be to remain the performance of existing systems without the need to retrain them from the beginning

"After supervised learning — (Lifelong) Transfer Learning will be the next driver of ML commercial success"

(Andrew Ng, in NIPS 2016)



Contents

01 Introduction

02 Prerequiste : Knowledge Transfer

03 Challenge : Catastrophic Forgetting

04 Solution : Continual Lifelong Learning

05 Conclusion

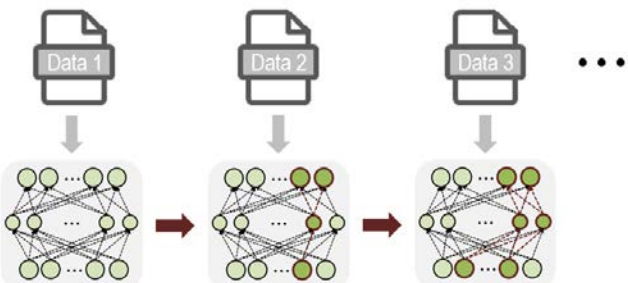
Understanding Knowledge Transfer

* Leverage features, weights from previously trained model

- ✓ After the model goes through the several steps of knowledge transfer, it generalizes its knowledge (feature representations, weights) better for the subsequent problem spaces

Isolated,
One task per one NN

<Traditional Learning Scheme>



<Transfer Learning Scheme >

Formal Definition of Knowledge Transfer

✓ Problem space is defined with the following factors:

① Domain $D = \{\chi,\ P(X)\}$: two-element tuple consisting of
- feature space χ
- marginal probability $P(X),\ X = \{x_1, \ldots, x_n\},\ x_i \in \chi$

② Task $T = \{\gamma,\ P(Y|X)\} = \{\gamma,\ \eta\}$: two-element tuple consisting of
- label space γ
- objective function $P(Y|X) = \eta,\ Y = \{y_1, \ldots, y_n\},\ y_i \in \gamma, \eta(x_i) = y_i$

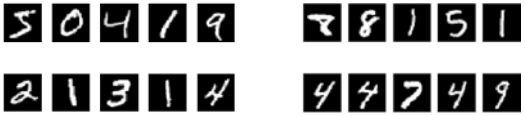
✓ Let the domain and task for t-th transfer phase as D_t , T_t respectively, then, the transfer process in $(t+1)$ -th problem space means:

"Learning the new objective function η_{t+1} with the information gained from D_{t+1}, and T_{t+1}"

Possible Scenarios in Knowledge Transfer

✓ If two domains are different, they may have different feature space (①) or different marginal distribution (②):

① $\chi_t \neq \chi_{t+1}$

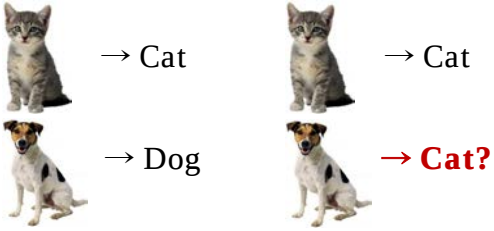


② $P(X_t) \neq P(X_{t+1})$

☞ Use Domain Adapatation

✓ If two tasks are different, they may have different label space (③) or different conditional distribution (④):

③ $\gamma_t \neq \gamma_{t+1}$



④ $P(Y_t|X_t) \neq P(Y_t|X_{t+1})$

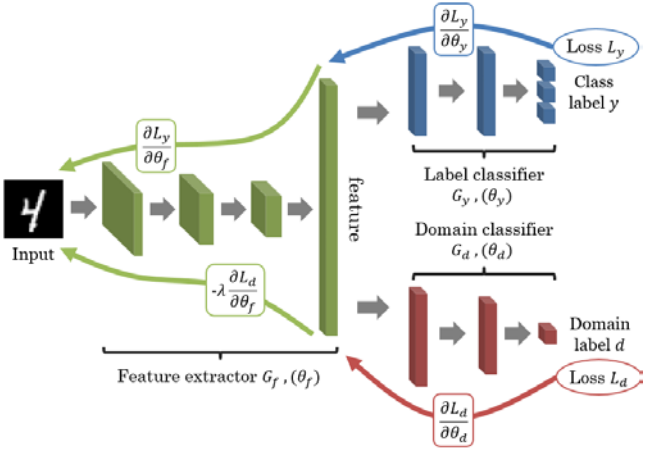
☞ Use Fine-Tuning

Transfer Strategies : Domain adaptaion

* Aligning the distribution of features across the two domains

✓ The main idea is to enjoin the network layer to learn a representation shared by both previous & current domain by

- Minimizing training classification error L_y
- Maximizing previous & current domain classification error L_d



▲Deep domain adaptation

Method	Previous	MNIST	SVHN
	Current	MNIST-M	MNIST
Train only previous domain		0.5225	0.5490
with Domain Adaptation		**0.7666**	**0.7385**
Train current domain individualy		0.9596	0.9942

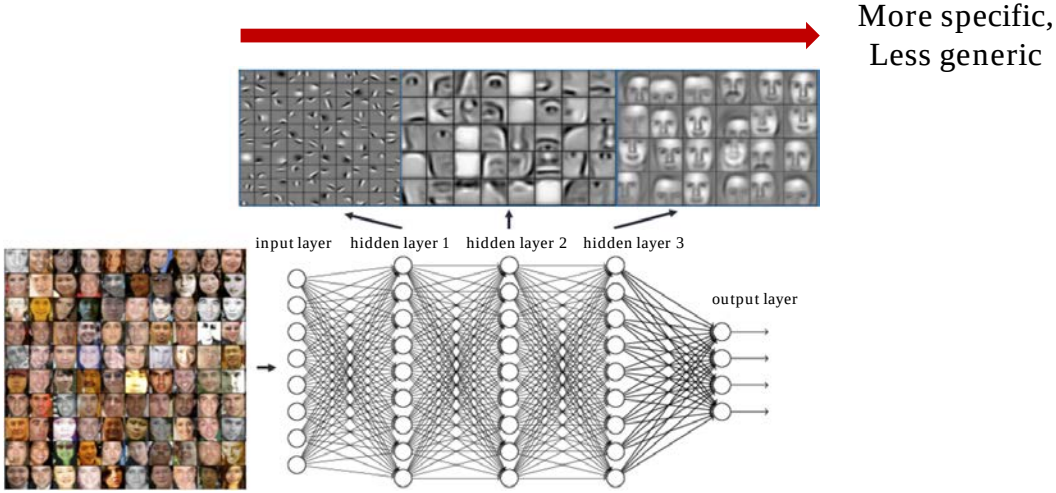
▲Digit classification accuracies after domain transfer

* Y. Ganin, et al. "Domain-Adversarial Training of Neural Networks", In Advances in Computer Vision and Pattern Recognition, 2017

Transfer Strategies : Fine-tuning

* Utilizing a part of pretrained models as feature extractors

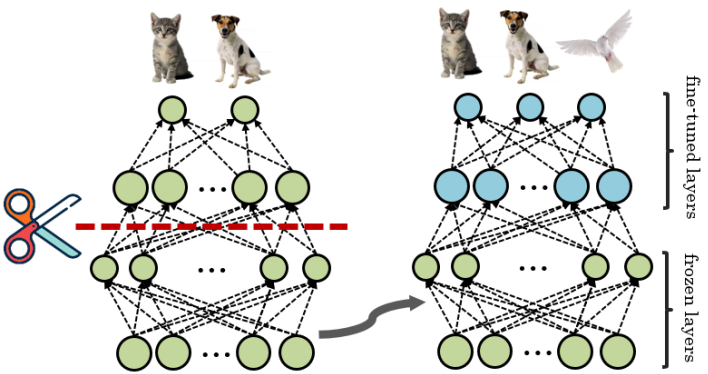
- ✓ The initial layers capture generic features, while the later ones focus more on the specific task at hand



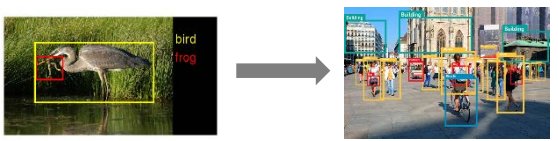
▲Neural network learn hierarchical feature representation

Transfer Strategies : Fine-tuning

✓ We can freeze certain layers by fixing their weights during the retraining and update the rest of them to suit our needs



▲Transfer with fine-tuning



ILSVRC 2013 dataset → New object classification

Object	Accuracy	Object	Accuracy
Hair	72.5	T-shirt	57.7
Glasses	55.6	Shorts	79.1
Hat	73.4	Jeans	78.1

▲Transfer on H3D dataset

* A. S. Razavian. et al. "CNN features off-the-shelf: An astounding baseline for recognition", In Proc. IEEE Conf. Computer Vision and Pattern Recognition, 2014

Contents

01 Introduction

02 Prerequiste : Knowledge Transfer

03 Challenge : Catastrophic Forgetting

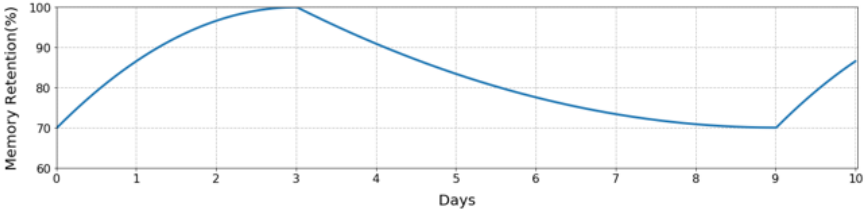
04 Solution : Continual Lifelong Learning

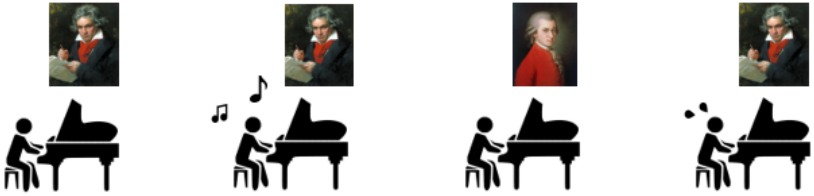
05 Conclusion

Forgetting What They Learned

* **Latest knowledge push out the old one when we do not try to retain it**

- ✓ Ideally, we want machines to learn in a sequential fashion, accumulating the knowledge learned in previous tasks
- ✓ But, previous knowledge is abruptly lost as information relevant to the current task is incorporated, which is well known as catastrophic forgetting

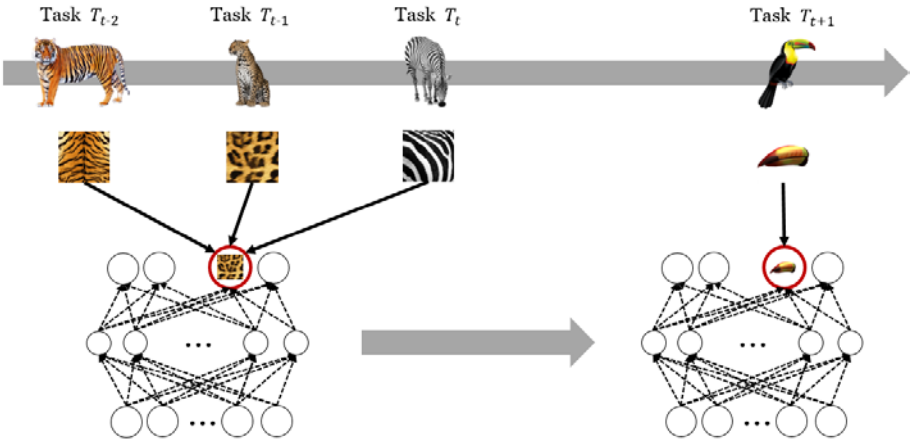




Semantic Drift

* **The original meaning of the features changes as they fit to later tasks**

✓ The interference of new feature map causes the semantic drift of certain hidden nodes, hence the model becomes more adaptable to the latest domain



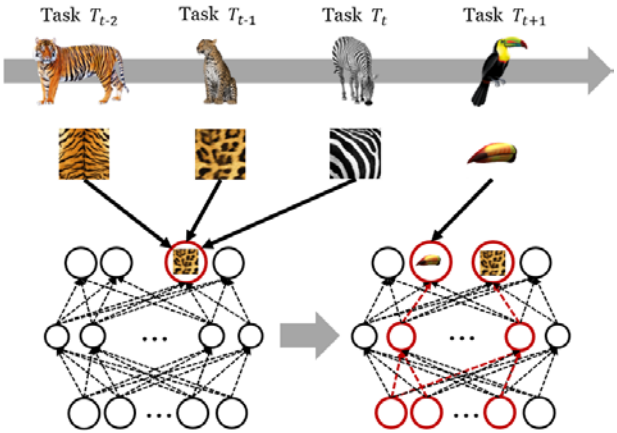
▲The weight associated with animal's skin could significantly change to classify the toucan

How to Combat the Catastrophic Forgetting

* **The fundamental role of the memory system is to determine the importance of the learned knowledge**

  ✓ People can recall stronger memories for a longer period; incorporating machine's knowledge should be done under the prioritized selectivity

Long-term Memory



▲The more impactful, repeated or explicit, the easier it will be absorbed in long-term memory

Contents

01 Introduction

02 Prerequiste : Knowledge Transfer

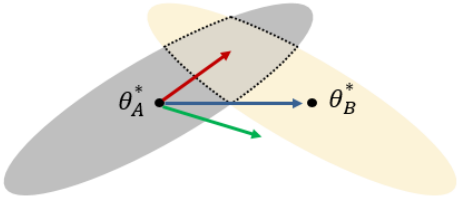
03 Challenge : Catastrophic Forgetting

04 Solution : Continual Lifelong Learning

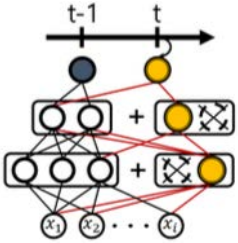
05 Conclusion

Research Trend

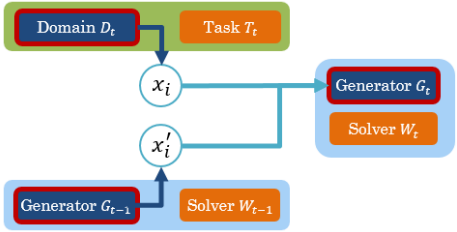
✓ The ongoing researches can be broken into 3 approaches:



Regularization approach



Structural approach



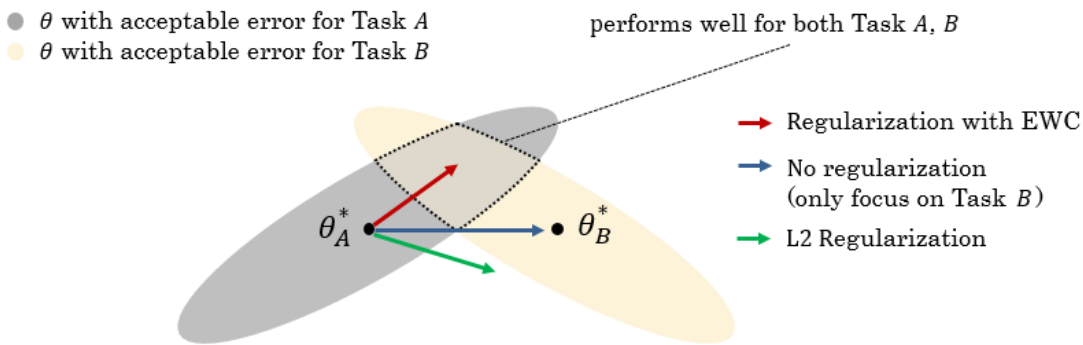
Generative approach

Regularization Approach : Elastic Weights Consolidation (EWC)

* EWC tries to converge to a point where both tasks have a low error

✓ We want to optimize the weight parameters $'\theta'$ to the intersection between the well-performed boundary for Task A and Task B, employing a regularization factor

① No regularization : the model will do better for Task B, not Task A
② L2 regularization : the model's performance is far from satisfactory
③ EWC : the model works well for both Task A and Task B



* J. Kirkpatrick. et al. "Overcoming catastrophic forgetting in neural networks", In Proc. National Academy of Sciences of the United States of America, 2017

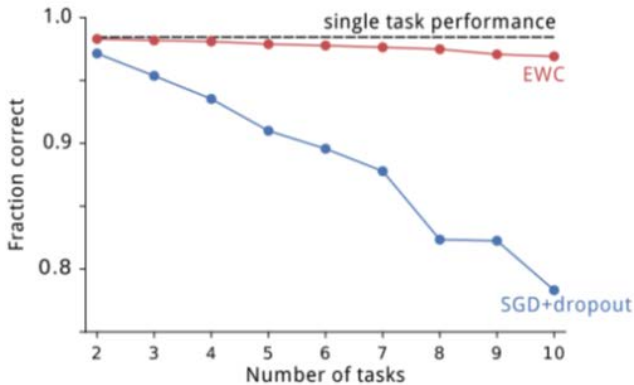
Regularization Approach : Elastic Weights Consolidation (EWC)

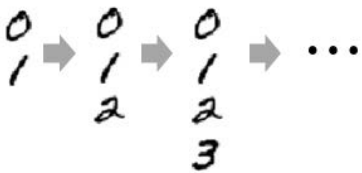
✓ The function $\mathcal{L}_{A,B}$ that we minimize in EWC is :

$$\mathcal{L}_{A,B}(\theta) = \mathcal{L}_B(\theta) + \underbrace{\frac{\lambda}{2} F(\theta - \theta_A^*)^2}_{\text{EWC Regularizer}}$$

Loss for task B only

where $'\lambda'$ is regularization parameter, and $'F'$ is Fisher information which estimate how fast we may forget task A depending on the direction





* J. Kirkpatrick. et al. "Overcoming catastrophic forgetting in neural networks", In Proc. National Academy of Sciences of the United States of America, 2017

Structural Approach : Dynamic Expandable Network (DEN)

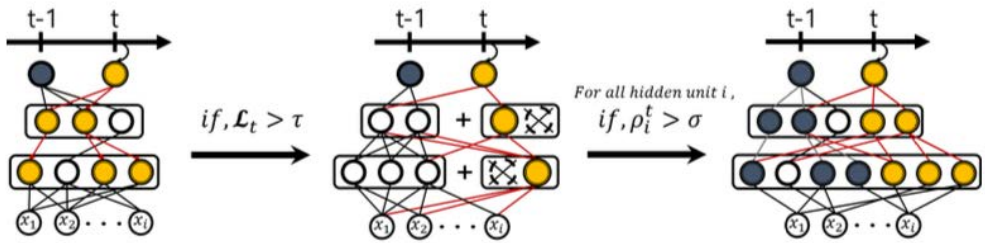
* Increase the storable capacity for the accommodation of new features

✓ DEN repeats the following 3 retraining steps:

① Selective retraining : figure out the old neurons that are relevant to the new task, and retrain them → compute the new loss $\mathcal{L}_t$

② Dynamic network expansion : if $\mathcal{L}_t$ is larger than threshold value $'\tau'$, then add more neurons and keep retraining → compute the displacement of i-th weight p_i^t

③ Network split/duplication : if p_i^t is larger than parameter $'\sigma'$, duplicate the corresponding weights and retrain them, while keeping the old weights fixed



* J. Yoon. et al. "Lifelong learning with dynamically expandable networks", In International Conference on Learning Representations, 2018

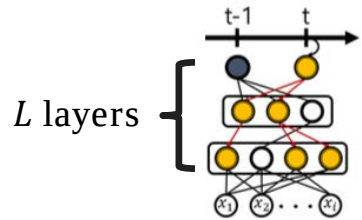
Structural Approach : Dynamic Expandable Network (DEN)

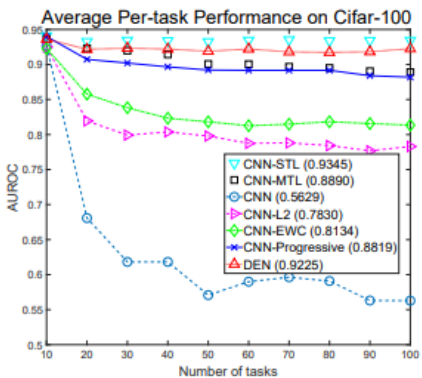
✓ But, how to select the subnetwork S ?

Algorithm 1 Selective Retraining

Input: previous weight configuration θ_{t-1}
Output: new weight configuration θ_t

1: Fine-tune last layer, i.e., $\min_{\theta_t^L} \mathcal{L}(\theta_t^L; \theta_t^{1:L-1}) + \lambda \sum_i ||\theta_t^i||_1$
2: Initialize $S = \{\text{nodes in output layer}\}$
3: Add node p to S, if there exists some nodes $q \in S$ such that $\theta_t^{L,pq} \neq 0$
4: **for** $l = L-1, ..., 1$ **do**
5: Add node p to S if there exists some nodes $q \in S$ such that $\theta_{t-1}^{l,pq} \neq 0$
6: **end for**
7: Solve $\min_{\theta_{t,S}} \mathcal{L}(\theta_{t,S}) + \lambda \sum_i ||\theta_{t,S}^i||_2$





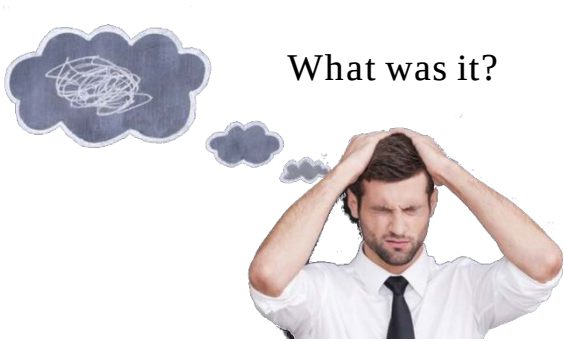
Cifar-100 Dataset

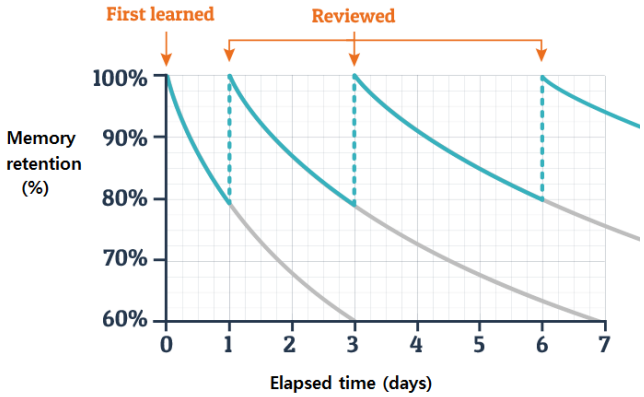
* J. Yoon. et al. "Lifelong learning with dynamically expandable networks", In International Conference on Learning Representations, 2018

Generative Approach : Memory Replay

* **Frequently encountered tasks tend to survive much longer in memory**

✓ Information is easier to recall when it's built upon things you already know, so staging frequent memory interventions solidifies the information

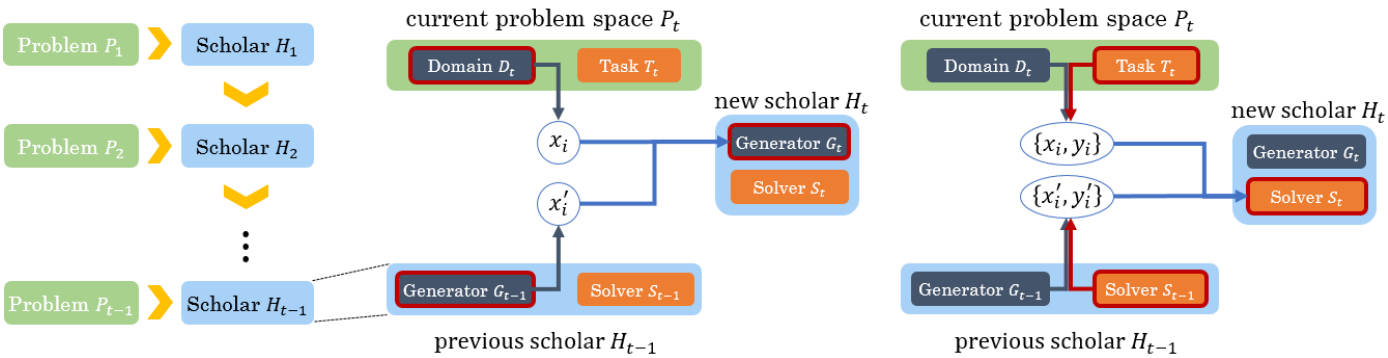




Generative Approach : Memory Replay

✓ Complementary (*new memory* ↔ *recall*) learning system can be implemented with a combination of a generator and a solver:

- Scholar H_t : tuple consisting of generator and solver $\{G_t, S_t\}$
- Generator G_t : generative model that produces manipulated replay x_i' mimics an original sample x_i from domain D_t
- Solver S_t : task solving model for objective function $'\eta'$ satisfying both $\eta(x_i) = y_i$ and $\eta(x_i') = y_i'$



* H. Shin. et al. "Continual learning with deep generative replay", In Advances in Neural Information Processing Systems (NIPS), 2017

Generative Approach : Memory Replay

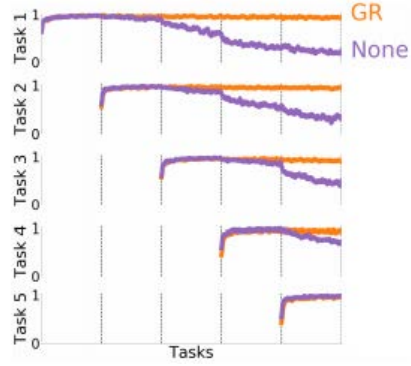
✓ The overall training process can be described as:

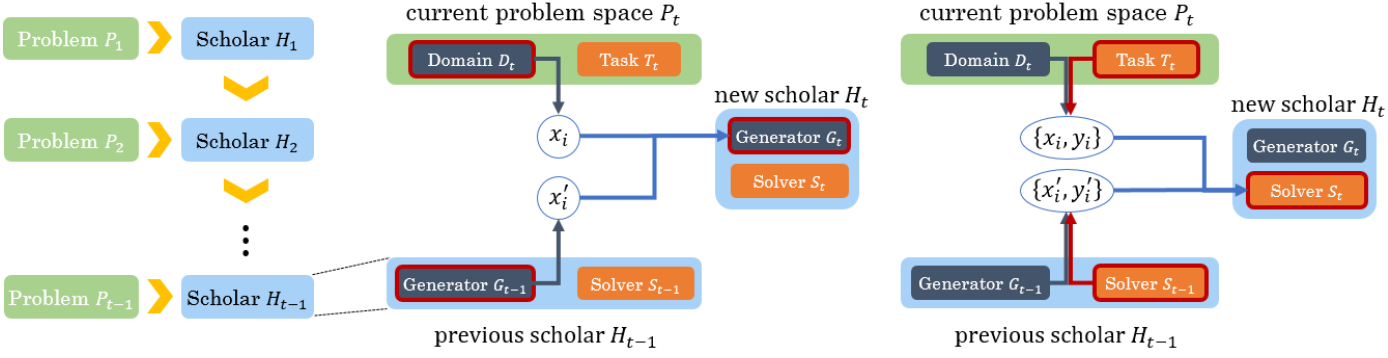
① Initially ($t = 1$), S_1 is run on the only x, and G_1 learns to mimic x

② From the second steps ($t > 1$), G_t learns both current data x, and x' generated by previous generator G_{t-1}

③ Previous solver S_{t-1} make prediction y' for the x'

④ Current solver S_t learns both $\{x, y\}$ and $\{x', y'\}$





* H. Shin. et al. "Continual learning with deep generative replay", In Advances in Neural Information Processing Systems (NIPS), 2017

Contents

01 Introduction

02 Prerequiste : Knowledge Transfer

03 Challenge : Catastrophic Forgetting

04 Solution : Continual Lifelong Learning

05 Conclusion

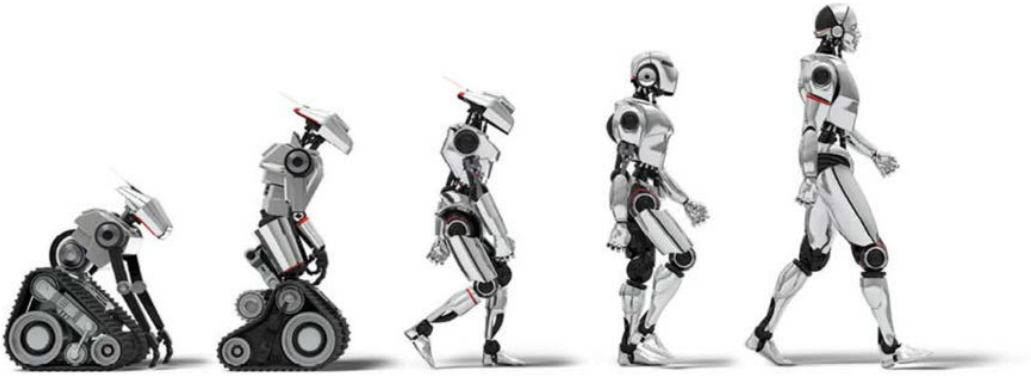
Conclusion

* **Never stop learning, because life never stops teaching**

  ✓ Without a way to avert catastrophic forgetting, a learner needs to be presented with all the training data at once, and cannot be retrained on new data thereafter

* **The solution is almost there : Lifelong learning**

  ✓ Lifelong learning is definitely going to be one of the key drivers for deep learning success in mainstream adoption in the industry



Thank You.